Enhanced Cervical Cancer Cell Detection Using CNN Transfer Learning with Strategic Layer Freezing

Felicia Audrey Tanujaya
Computer Science Department, School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
felicia.tanujaya002@binus.ac.id

Audrey Theodora Phang
Computer Science Department, School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
audrey.phang@binus.ac.id

Kelly Natalia
Computer Science Department, School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
kelly.natalia@binus.ac.id

Dave Christian Thio
Bioinformatics and Data Science Research Center
Bina Nusantara University
Jakarta, Indonesia
dave.thio@binus.ac.id

Gregorius Natanael Elwirehardja
Computer Science Department, School of Computer Science
Bina Nusantara University
Jakarta, Indonesia
gregorius.elwirehardja@binus.ac.id

Bens Pardamean
Computer Science Department, BINUS Graduate Program - Master of Computer Science
Bina Nusantara University
Jakarta, Indonesia
bpardamean@binus.edu

***Abstract*—Cervical cancer remains a leading cause of cancer-related mortality among women, particularly in regions with limited access to early and accurate diagnostics. This study explores the use of deep learning for automated cervical cell classification in resource-constrained settings. The SIPaKMeD dataset, comprising 4,049 expert-labeled single-cell images in five morphological classes, was selected for its quality and representation of real-world cytology variability. Data were partitioned into training, validation, and test sets through stratified sampling to preserve class balance. Four Convolutional Neural Network (CNN) architectures were evaluated in base and fine-tuned configurations, chosen for their proven ability to extract spatial features from medical images. Transfer learning from ImageNet weights addressed limited medical image availability, with early layers frozen to retain general features and deeper layers adapted for domain-specific patterns. Fine-tuned ResNet50V2 achieved the highest F1-score of 0.9154, with consistent performance gains across all models and metrics. These results demonstrate the potential of combining transfer learning with selective layer freezing to enhance cervical cancer detection in limited-data scenarios, offering a practical pathway for AI-assisted screening in low-resource healthcare environments.**

***Index Terms*—cervical cancer, deep learning, convolutional neural networks, transfer learning**

I. Introduction

Cervical cancer is a major contributor to cancer-related mortality among women worldwide, particularly in Low and Middle Income Countries (LMICs) where access to preventive screening and healthcare services is severely limited [1]. In 2022, over 660,000 new cases and approximately 350,000 fatalities were reported globally, with the highest burden concentrated in transitioning countries across Asia and Africa [2]. Persistent infection with oncogenic human papillomaviruses (HPVs), particularly types 16 and 18, is responsible for approximately 70% of cervical cancer cases [3].

Despite the availability of effective vaccines and cytology-based screening methods, LMICs continue to face significant challenges due to infrastructure limitations, insufficient numbers of qualified trained medical professional, and sociocultural barriers to participation [4]– [6]. Traditional Pap smear analysis remains time-consuming, subjective, and prone to diagnostic variability, leading to growing interest in artificial intelligence (AI)-driven alternatives.

Deep Convolutional Neural Networks (CNNs), a cornerstone of modern medical image classification, have demonstrated superior performance by automatically learning hierarchical image features, as shown in reviews such as Cai *et al.* on medical imaging applications [7]. Recent studies leveraging transfer learning with architectures like ResNet-50, MobileNet V2, and DenseNet121 have achieved classification accuracies above 98% on public cervical cytology datasets such as Herlev and SIPaKMeD [8]– [10]. Alongside these supervised transfer learning methods, other advanced approaches such as self-supervised learning are also being explored to tackle the classification of pap smear cells [11].

However, the specific effects of layer-freezing methods remain underexplored in cervical cytology models. In this approach, initial layers are frozen to preserve learned representations, while deeper layers are fine-tuned for domain-specific adaptation. This gap limits potential accuracy gains in early and reliable cervical cancer detection, especially in data-scarce settings. Previous studies in other medical imaging domains show that such partial network adaptation can improve performance on limited datasets [12], [13].

This study aims to investigate the effectiveness of deep learning, specifically Convolutional Neural Networks (CNNs) for the automated classification of cervical cancer cells using the SIPAKMED dataset. We evaluate four CNN models: EfficientNetB0, ConvNeXt Tiny, ResNet50V2 and EfficientNetV2-Small. These architectures were chosen to span from lightweight models for efficiency to deep networks with high representational capacity, allowing a balanced evaluation of performance trade-offs in medical image classification. Each model is assessed in two configurations: a base model with frozen layers and a fine-tuned version where selected layers are unfrozen for further adaptation.

II. Methodology

A. Dataset

This study used the publicly available SIPaKMeD dataset [14], which comprises 4,049 isolated cervical cell images manually extracted from 966 cluster images of Pap smear slides. The dataset is publicly available and provides a well-annotated benchmark for cervical cell classification tasks. Each single-cell image is categorized into one of five classes representing both normal and pathological cervical cell types: superficial–intermediate, parabasal, koilocytotic, dyskeratotic, and metaplastic.

The dataset was split into training, validation, and test subsets using stratified sampling to preserve class balance. The detailed distribution of images per class across the three subsets is presented in Table I.

TABLE I
Class Distribution Across Dataset Splits

Class	Number of Samples		
	Train	*Validation*	*Test*
Dyskeratotic	648	65	100
Koilocytotic	656	69	100
Metaplastic	624	69	100
Parabasal	612	75	100
Superficial-Intermediate	655	76	100
Total	3195	354	500

Overall, the class distribution is approximately balanced across all subsets, ensuring fair evaluation during training and testing phases.

B. Data Augmentation

All images were resized to 224×224 pixels to match the input requirements of the convolutional neural networks used in this study. Following resize, images were normalized through per-image standardization, which involves subtracting the mean pixel value and dividing it by the standard deviation to reduce the impact of lighting inconsistencies. Subsequently, random horizontal flips were introduced to account for possible variations in cell orientation. Additionally, brightness and contrast were adjusted randomly within defined ranges to simulate the variability in staining and illumination commonly observed in cervical cytology images. The contrast varied between 0.7 and 1.3 times the original level, while brightness was modified by a maximum delta of 0.2, as shown in Fig. 1.

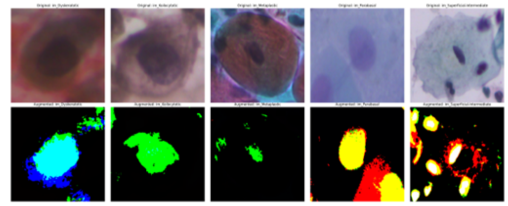

Fig. 1. Original and augmented image samples.

C. Model Architecture

This study used four deep learning architectures: EfficientNetB0, ConvNeXt Tiny, ResNet50V2, and EfficientNetV2-Small, all initialized with pre-trained weights. At the initial stage, all layers were frozen so the models could act as fixed feature extractors, making use of general patterns like edges and shapes learned from large datasets [15]. EfficientNetB0 was chosen for its strong performance and efficient use of computational resources, striking a good balance between depth and width [16]. ConvNeXt Tiny, inspired by vision transformers, was chosen for its ability to capture general features effectively while maintaining computational efficiency [17]. ResNet50V2 was included due to its use of residual connections, which mitigate the vanishing gradient problem and facilitate deep network training [18]. Lastly, EfficientNetV2-Small was selected for its improved MBConv layers, offering high accuracy with efficient resource use.

Next, we performed fine-tuning by unfreezing some of the last blocks of each model. We fine-tuned blocks 1-3 while keeping the earlier layers frozen, as they capture general visual features that work well across different datasets. This approach helps to reduce the risk of overfitting and improves training efficiency on our dataset [19]. The methodology architecture can be viewed at Fig. 2.

D. Model Training and Hyperparameter Optimization

The base setup for all four models followed a consistent approach using transfer learning with pre-trained ImageNet weights. The convolutional layers were frozen to prevent updates during the initial training phase, ensuring that the pre-learned features remain intact. This approach reduces the risk of overfitting, particularly given the limited size of the cervical cell dataset. A new classifier was appended to the frozen base, consisting of a dense layer with 128 units and ReLU activation, followed by a dropout layer with a rate of 0.3 to reduce overfitting. The final layer used a softmax activation with 5 units to output the class probabilities. During training, only the newly added classifier layers were trained, while the convolutional layers remained fixed.

For the learning rate, we used 1e-4 because it provided a good trade-off between speed and stability, avoiding issues like

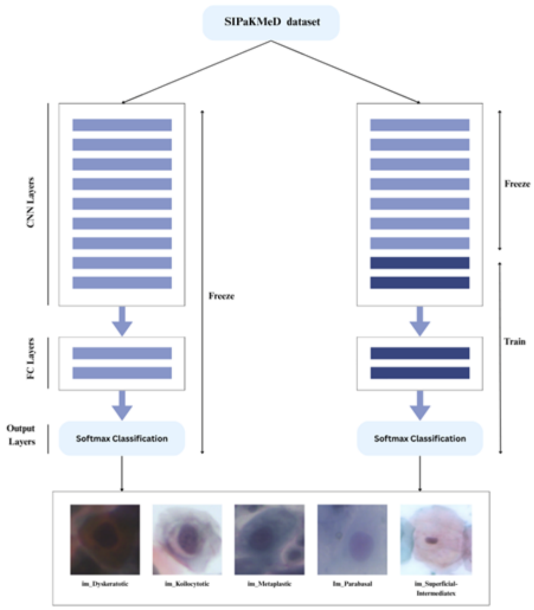

Fig. 2. Deep learning architecture for SipakMed dataset classification using frozen CNN layers.

dying ReLU where neurons stop activating. For the optimizer, we used Adam optimizer because of its adaptive learning capabilities, and also early stopping with a patience of 10 was applied to reduce overfitting and restore the model to its best-performing weights during training.

TABLE II
OPTIMIZED HYPERPARAMETER SETTINGS FOR BASE MODELS

Hyperparameter	**Model**			
	EfficientNet B0	***ConvNeXt Tiny***	***ResNet50V2***	***EfficientNetV2-Small***
Learning Rate	1e-05	1e-04	5e-05	5e-05
L2 Regularization	0.001	5e-05	5e-05	0.001
Unfrozen Blocks	block5, block6, block7	stage3	conv4, conv5	block4, block5, block6

Table II presents the final hyperparameter settings used for each model after the fine-tuning process. For each hyperparameter, we experimented with three different values. Specifically, we tested learning rates of 1e-4, 1e-5, and 5e-5, and L2 regularization values of 1e-3, 1e-4, and 5e-5. As part of fine-tuning, we also unfroze up to three deeper blocks in each model to allow learning of more task-specific features while keeping early layers frozen to retain general representations and avoid overfitting. The values shown in the table are those that resulted in the best performance for each model based on validation results.

E. Evaluation Metrics

The performance of the models was assessed using several standard evaluation metrics: accuracy, precision, recall, and F1-score. In the context of this study, where the dataset includes a range of normal and pathological cervical cell types, it is important to assess both the overall classification accuracy and the model's capacity to accurately identify each class.

Given the implications of false positives and false negatives in medical diagnostics, F1-score is adopted as the primary evaluation metric, as it balances precision and recall by considering both types of errors equally. The F1-score (1) is calculated using the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{1}$$

Additionally, we also used the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) to further evaluate model performance. The ROC curve shows the True Positive Rate (TPR) (2) against the False Positive Rate (FPR) (3) at different thresholds, where:

$$TPR = \frac{TP}{TP + FN} \tag{2}$$

$$FPR = \frac{FP}{FP + TN} \tag{3}$$

The AUC represents the area under the ROC curve and provides a single value summarizing the model's ability to distinguish between positive and negative cases. A higher AUC indicates that the model performs better at differentiating between classes, offering a clear summary of its classification capability [20].

III. RESULT AND DISCUSSION

A. Result

TABLE III
EVALUATION METRICS OF BASE MODELS

Algorithm	**Evaluation Metrics**			
	Precision	***Recall***	***F1-Score***	***Accuracy***
EfficientNetB0	0.5233	0.5080	0.4439	0.5080
ConvNeXt Tiny	**0.8501**	**0.8440**	**0.8440**	**0.8440**
ResNet50V2	0.0402	0.1940	0.0667	0.1940
EfficientNetV2-Small	0.1863	0.2200	0.1950	0.2200

ResNet50V2 achieved the highest performance after fine-tuning, with an F1-Score of 0.9154, as shown in Table IV Despite its weaker baseline performance, fine-tuning enabled the model to detect positive cases with high precision and recall. In a clinical context, this F1-Score reflects strong diagnostic reliability, minimizing false positives and false negatives, which is crucial to avoid delayed treatment or unnecessary interventions [21].

For baseline results presented in Table III, ConvNeXt Tiny led with an F1-Score and accuracy of 0.8440, showing strong ability to correctly identify positive cases even without fine-tuning. EfficientNetB0 performed moderately (F1-Score 0.4439), while ResNet50V2 and EfficientNetV2-Small performed poorly, missing many correct samples. ConvNeXt Tiny's strong baseline performance is likely due to its efficient convolutional design and pretrained features.

Fine-tuning improved all models. ConvNeXt Tiny showed only slight gains, whereas EfficientNetB0, EfficientNetV2-Small, and ResNet50V2 improved substantially. ResNet50V2's deep residual architecture allowed effective adaptation to dataset-specific patterns, making it the best-performing model after fine-tuning. This highlights that while pretrained features are important, deeper and flexible architectures can achieve superior results when fine-tuned.

TABLE IV
EVALUATION METRICS OF FINE-TUNED MODELS

Algorithm	Evaluation Metrics			
	Precision	*Recall*	*F1-Score*	*Accuracy*
EfficientNet B0	0.9007	0.8860	0.8882	0.8860
ConvNeXt Tiny	0.8531	0.8480	0.8480	0.8480
ResNet50V2	**0.9194**	**0.9140**	**0.9154**	**0.9140**
EfficientNetV2-Small	0.9078	0.9000	0.9012	0.9000

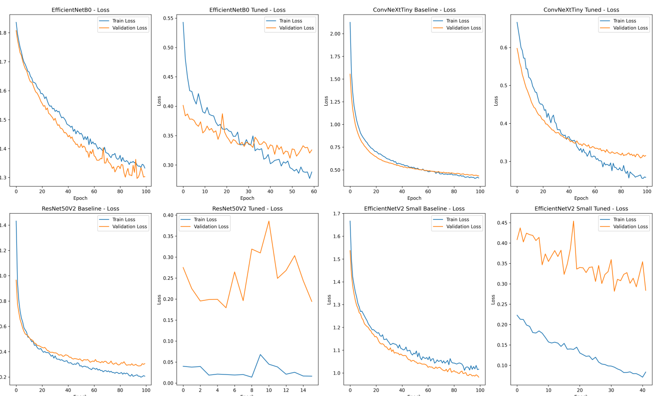

Fig. 3. Loss curves during training and validation.

The loss curves in Fig. 3 illustrate training and validation performance across models. EfficientNetB0 shows steady training convergence, with fine-tuning improving validation loss consistency, suggesting better generalization. ConvNeXt Tiny and ResNet50V2 also show clear loss reductions. In contrast, EfficientNetV2-Small exhibits notable fluctuations in validation loss, particularly after fine-tuning, which may stem from its limited capacity or sensitivity to hyperparameters and could affect prediction consistency. Overall, fine-tuned models consistently outperform their base versions.

Fig. 4 demonstrate significant improvements in the ROC curves for classification performance after fine-tuning the models. For the base models, EfficientNetB0 and ConvNeXt Tiny perform reasonably well with AUC scores ranging from 0.7100 to 0.9891 across different classes. After fine-tuning, all models show substantial performance gains, with ConvNeXt Tiny reaching an AUC of 0.9942 for class 4, while EfficientNetB0 achieves a remarkable AUC of 0.9987 for class 0. ResNet50V2 and EfficientNetV2-Small, which initially showed poor performance, exhibit significant improvements post-tuning, with EfficientNetV2-Small achieving the highest AUC of 0.9994 for class 0.

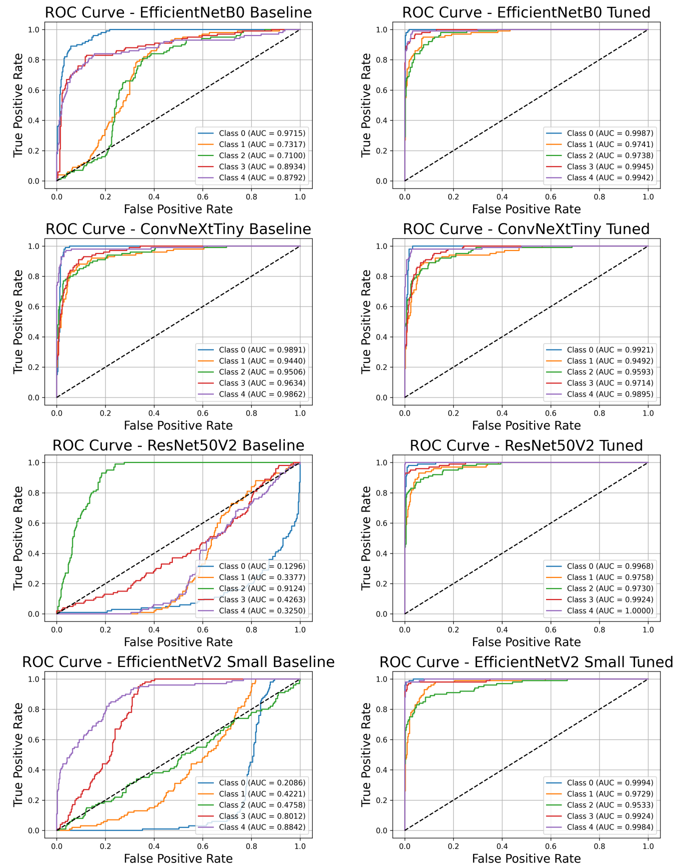

Fig. 4. ROC curve for all models.

Following the quantitative evaluation, a qualitative assessment was conducted on the fine-tuned ResNet50V2 model, which achieved the highest F1-Score 0.9154 among all tested architectures. Fig. 5 illustrates representative test samples correctly classified by the model, with high confidence scores across varying cell classes, including metaplastic, koilocytotic, parabasal, and superficial-intermediate. Each prediction is accompanied by its corresponding ground truth label and softmax probability score, all reaching 1.00, indicating strong model certainty.

B. Discussion

This study highlights the effectiveness of combining transfer learning with strategic layer freezing in improving cervical cell classification on the SIPAKMED dataset. Among all tested architectures, ConvNeXt Tiny demonstrated the strongest performance in its base (frozen) state, suggesting that its pre-trained weights already captured highly transferable features suited for this task. In contrast, ResNet50V2 and EfficientNetV2-Small, which initially performed poorly, showed substantial gains after fine-tuning, with ResNet50V2 ultimately outperforming all other models post-tuning.

This pattern suggests that fine-tuning effectiveness depends on how well pre-trained features align with the target data. ConvNeXt Tiny's hybrid design yields more general-

Fig. 5. Sample predictions from fine-tuned ResNet50V2.

izable representations that minimize adaptation needs, while ResNet50V2 required deeper layer unfreezing to capture cytological features, indicating that layer-wise adaptation is crucial for models with less optimal initial features.

TABLE V
RESULTS OBTAINED IN PREVIOUS STUDIES

Study	Model(s)	Dataset(s)	Task	Transfer Learning / Fine-tuning Strategy	Performance Metric
[22]	Cerviformer	SIPaKMeD, Herlev	2-class, 3-class	Asymmetric attention-based model	Accuracy: 0.9370 (Sipakmed), 0.9470 (Herlev)
[23]	SqueezeNet	SIPaKMeD	2-class	Median Filter Based Preprocessing	Precision: 0.970
[9]	Inception-ResNet-V2	SIPaKMeD	5-class	LR fine-tuning, fuzzy ensemble	F1-Score: 0.9639
[8]	ResNet152	SIPaKMeD	2-class	DL feature extraction, ML prediction	Precision: 0.9521, Recall: 0.9512
This study	**ResNet50V2**	**SIPaKMeD**	**5-class**	**DL feature extraction, strategic freezing & unfreezing, hyperparameter tuning**	**Accuracy: 0.9140, F1-Score: 0.9194**

A comparative overview is provided in Table V. ResNet50V2 achieved an F1-score of 0.9154, showing a significant improvement in this study, but it is numerically lower than some accuracies reported in other SIPaKMeD studies. These differences arise from several factors. First, this study reports F1-score, a more conservative metric than accuracy in medical diagnostics. Second, higher accuracies in the literature often come from simpler 2- or 3-class tasks, which are less complex than the 5-class classification here, as seen with CerviFormer achieving 0.937 accuracy for a 3-class task [19]. Third, some high-performing works use ensemble methods, which typically outperform individual models [9].

Therefore, the primary contribution of this paper lies not in setting a new accuracy benchmark, but in providing a systematic and in-depth exploration of strategic layer freezing and its differential impact across a diverse set of modern CNN architectures for cervical cell classification. While our model has not yet reached full convergence, the consistent performance improvements observed after fine-tuning suggest that further optimization and extended training could yield even better results.

Our study has several limitations, mainly due to relying solely on the SIPaKMeD dataset. Its pre-cropped, high-quality images may lead to overfitting and limit generalizability, as real-world clinical images can vary in quality and consistency. The dataset may also contain biases in class distribution or sampling, affecting model predictions. Future work should include additional datasets from diverse sources to improve variability, robustness, and generalization.

Moreover, Future research should explore explainability techniques, such as attention maps or Grad-CAM, to provide clear visual insights into which features the model focuses on when classifying cell types [24]. These approaches can enhance trust and interpretability for medical practitioners, help ensure that model predictions align with domain expertise, and allow experts to verify that classification results are medically reasonable and reliable [25].

Overall, our study demonstrates that combining transfer learning with strategic layer freezing enhances cervical cell classification on the SIPaKMeD dataset. While ConvNeXt Tiny performs strongly with its pretrained features, ResNet50V2 gains the most from fine-tuning, underscoring the importance of layer-wise adaptation for less aligned models. These findings emphasize that architecture choice and fine-tuning strategy are critical for maximizing performance and offer practical guidance for deep learning in cytological diagnostics.

IV. CONCLUSION

This study assesses the impact of transfer learning with strategic layer freezing on cervical cancer cell classification using the SIPaKMeD dataset. Among the models evaluated, ConvNeXt Tiny achieved the highest overall performance with an F1 score of 0.8480, while ResNet50V2 achieved the highest individual class F1-score of 0.9154 for the parabasal cell type. These results demonstrate that applying strategic layer freezing can enhance model generalization and training efficiency, particularly in limited-data scenarios, and may improve AI-driven cervical cancer screening tools by reducing diagnostic time and interobserver variability in clinical workflows. However, this study is limited by the use of a single high-quality dataset, which may not fully represent real-world variability and introduces a risk of overfitting. Future studies may explores multi-dataset experiments, domain adaptation techniques, and hybrid CNN–attention architectures to improve robustness and clinical applicability. Overall, the findings highlight the potential of combining transfer learning with strategic layer freezing to advance AI-assisted cervical cancer diagnostics, offering a pathway toward more accurate and efficient clinical screening solutions.

Acknowledgment

The authors would like to express their sincere gratitude to Bioinformatics & Data Science Reasearch Center (BDSRC) at BINUS University for providing invaluable supervision and technical guidance throughout this study. This study was conducted as part of a collaboration between the BINUS University's Data Science Club (DSC) and BDSRC.

Author Contributions

Felicia Audrey Tanujaya: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Writing - Original Draft. Audrey Theodora Phang: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing - Original Draft. Kelly Natalia: Conceptualization, Formal Analysis, Writing - Original Draft. Dave Christian Thio: Conceptualization, Writing - Original Draft, Project Administration. Gregorius Natanael Elwirehardja: Conceptualization, Writing - Review & Editing, Supervision, Project Administration. Bens Pardamean: Writing - Review & Editing, Supervision.

References

[1] R. Hull *et al.*, "Cervical cancer in low and middle-income countries," *Oncol. Lett.*, vol. 20, no. 3, pp. 2058–2074, Sep. 2020.

[2] J. Wu *et al.*, "Global burden of cervical cancer: current estimates, temporal trend and future projections based on the GLOBOCAN 2022," *J. Natl. Cancer Cent.*, vol. 5, no. 3, pp. 322–329, 2025, doi: 10.1016/j.jncc.2024.11.006.

[3] A. A. McBride *et al.*, "Human papillomaviruses: diversity, infection and host interactions," *Nat. Rev. Dis. Primers*, vol. 7, no. 1, pp. 1–20, Mar. 2021, doi: 10.1038/s41579-021-00617-5.

[4] O. Farajimakin *et al.*, "Barriers to cervical cancer screening: A systematic review," *Cureus*, vol. 16, no. 7, Jul. 2024.

[5] A. Srinath *et al.*, "Barriers to cervical cancer and breast cancer screening uptake in low- and middle-income countries: a systematic review," *Health Policy Plan.*, vol. 38, pp. 509–527, 2023.

[6] I. Rosato *et al.*, "Adherence to cervical cancer screening programs in migrant populations: a systematic review and meta-analysis," *Int. J. Environ. Res. Public Health*, vol. 20, no. 3, 2023.

[7] L. Cai, J. Gao, and D. Zhao, "A review of the application of deep learning in medical image classification and segmentation," *Ann. Transl. Med.*, vol. 8, no. 11, p. 713, Jun. 2020, doi: 10.21037/atm.2020.02.44.

[8] S. K. Mathivanan, D. Francis, S. Srinivasan, V. Khatavkar, P. Karthikeyan, and M. A. Shah, "Enhancing cervical cancer detection and robust classification through a fusion of deep learning models," *Sci. Rep.*, vol. 14, no. 1, Art. no. 10812, 2024, doi: 10.1038/s41598-024-61063-w.

[9] R. Pramanik, M. Biswas, S. Sen, L. A. de Souza Júnior, J. P. Papa, and R. Sarkar, "A fuzzy distance-based ensemble of deep models for cervical cancer detection," *Comput. Methods Programs Biomed.*, vol. 219, Art. no. 106776, 2022, doi: 10.1016/j.cmpb.2022.106776.

[10] Y. Zhang, C. Ning, and W. Yang, "An automatic cervical cell classification model based on improved DenseNet121," *Scientific Reports*, vol. 15, no. 1, Jan. 2025.

[11] E. Selvano, G. Elwirehardja, and B. Pardamean, "SELF-SUPERVISED MODEL IN MULTI-TASK LEARNING FOR CLASSIFYING PAP SMEAR CELLS," *ICIC Express Lett.*, vol. 15, pp. 1193–1199, Nov. 2024, doi: 10.24507/icicelb.15.11.1193.

[12] A. Davila, J. Colan, and Y. Hasegawa, "Comparison of fine-tuning strategies for transfer learning in medical image classification," *Image Vis. Comput.*, vol. 145, Apr. 2024, Art. no. 105012, doi: 10.1016/j.imavis.2024.105012.

[13] H. E. Kim *et al.*, "Transfer learning for medical image classification: a literature review," *BMC Med. Imaging*, vol. 22, no. 69, 2022.

[14] M. E. Plissiti, P. Dimitrakopoulos, G. Sfikas, C. Nikou, O. Krikoni, and A. Charchanti, "Sipakmed: A new dataset for feature and image based classification of normal and pathological cervical cells in Pap smear images," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Athens, Greece, 2018, pp. 3144–3148, doi: 10.1109/ICIP.2018.8451588.

[15] M. Iman, H. R. Arabnia, and K. Rasheed, "A review of deep transfer learning and recent advancements," *Technologies*, vol. 11, no. 2, p. 40, Mar. 2023, doi: 10.3390/technologies11020040.

[16] M. Alruwaili and M. Mohamed, "An integrated deep learning model with EfficientNet and ResNet for accurate multi-class skin disease classification," *Diagnostics*, vol. 15, no. 5, p. 551, 2025, doi: 10.3390/diagnostics15050551.

[17] D. Setiawan, A. S. Karnyoto, I. Intan, and B. Pardamean, "ConvNeXt Model for Breast Cancer Image Classification," in *Proc. 2024 6th Int. Conf. Cybernetics Intell. Syst. (ICORIS)*, Surakarta, Indonesia, Nov. 29–30, 2024, pp. 1–5, doi: 10.1109/ICORIS63540.2024.10903832.

[18] L. Borawar and R. Kaur, "ResNet: Solving vanishing gradient in deep networks," in *Adv. Artif. Intell. Data Eng.*, Springer, vol. 978-981-19-8824-0, pp. 235–247, 2023, doi: 10.1007/978-981-19-8825-7_21.

[19] T. E. Moudden, M. Amnai, A. Choukri, Y. Fakhri, and G. Noreddine, "New unfreezing strategy of transfer learning in satellite imagery for mapping the diversity of slum areas: A case study in Kenitra city—Morocco," *Scientific African*, vol. 24, p. e02135, Jun. 2024, doi: 10.1016/j.sciaf.2024.e02135.

[20] H. H. Muljo *et al.*, "Handling severe data imbalance in chest X-Ray image classification with transfer learning using SwAV self-supervised pre-training," *Commun. Math. Biol. Neurosci.*, vol. 2023, no. 0, Art. ID 13, Jun. 2023, doi: 10.28919/cmbn/7526.

[21] T. B. Alakus and I. Turkoglu, "Comparison of deep learning approaches to predict COVID-19 infection," *Chaos, Solitons & Fractals*, vol. 140, p. 110120, Nov. 2020, doi: 10.1016/j.chaos.2020.110120.

[22] B. S. Deo, M. Pal, P. Panigarhi, and A. Pradhan, "CerviFormer: A Pap-smear based cervical cancer classification method using cross attention and latent transformer," arXiv preprint arXiv:2303.10222, Mar. 2023. [Online]. Available: https://arxiv.org/abs/2303.10222

[23] Z. Karapinar Senturk and S. Uzun, "An Improved Deep Learning Based Cervical Cancer Detection Using a Median Filter Based Preprocessing," *European Journal of Science and Technology*, 2022, doi: 10.31590/ejosat.1045538.

[24] Y. Guo, D. Chen, C. Bao, and Y. Luo, "Causal Attention-Based Lightweight and Efficient Cervical Cancer Cell Detection Model," 2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), vol. 9199, pp. 1104–1111, Dec. 2023, doi: 10.1109/bibm58861.2023.10385353.

[25] F. M. Talaat, S. A. Gamel, R. M. El-Balka, M. Shehata, and H. ZainEldin, "Grad-CAM Enabled Breast Cancer Classification with a 3D Inception-ResNet V2: Empowering Radiologists with Explainable Insights," Cancers, vol. 16, no. 21, p. 3668, Oct. 2024, doi: 10.3390/cancers16213668.